

Cost Overrun Prediction

COST OVERRUN · ML MODEL

An AI assistant and recommendation system that predicts the probability and magnitude of a project exceeding its sanctioned budget - updated monthly from live cost data. It flags the top cost-at-risk categories, identifies the specific drivers, and recommends corrective actions - giving commercial teams a 3-6 month window to act before losses lock in.

Input Dataset

— key fields per data section

Project Cost Records	Schedule & Progress	QA/QC & Defects	Procurement & Site	Project Profile
- Sanctioned budget by cost code - Actual spend - monthly - Committed spend (POs raised) - Forecast final cost (EAC) - Variation orders issued & approved - Rework cost by event	- Baseline vs actual schedule (P6/MSP) - Activity float & critical path flag - Baseline revision history - SPI & CPI per period - Labour headcount vs plan - Overtime & absenteeism rate	- Defect type, severity & location - Discovery date & construction stage - Responsible subcontractor - Resolution date & rework cost - Time-to-close (days)	- PO date vs actual delivery date - Material type, qty & supplier - Delay in days per PO - Site storage condition - Weather / non-productive days	- Project type & GFA (m²) - Floors & structural system - Contract type & value (INR) - Planned duration & city - Contractor class (A/B/C)

ML Model Outputs

Budget at risk	Predicted final cost vs sanctioned budget - in INR and %	Risk flag	RED (>10%) / AMBER (5–10%) / GREEN (<5%) signal
Cost breakdown	Risk by category: material / labour / rework / variation orders	Action trigger	Freeze PO / escalate VO / review subcontractor performance
Driver ranking	Top 3 overrun drivers ranked by cost contribution	Trend line	Projected overrun trajectory if no corrective action taken

Time Delay Prediction

TIME DELAY · ML MODEL

An AI assistant and recommendation system that predicts expected delay in days at each project milestone, identifies work packages carrying the highest schedule risk, and recommends resource reallocation and float adjustments - before the critical path breaks. Updated every reporting cycle with live schedule data.

Input Dataset

— key fields per data section

Baseline vs Actual Schedule	Weather & Site	QA/QC & Defects	Labour Deployment	Project Profile
- Activity ID, planned start & finish - Actual start & finish dates - Total float & critical path flag - Baseline revision history - Monthly P6 / MSP snapshot	- Daily rainfall (mm) at site - Temperature & wind speed - Non-productive days by cause - Planned vs actual working calendar - Monsoon window by city	- Defect type, severity & location - Discovery date & stage - Responsible subcontractor - Resolution date & rework cost - Time-to-close (days)	- Daily headcount by trade & work package - Planned headcount vs actual - Overtime hours & absenteeism rate - Skilled vs semi-skilled ratio - Subcontractor vs direct split	- Project type & GFA (m²) - Floors & structural system - Contract type & duration - Planned duration & city - Contractor class (A/B/C)

ML Model Outputs

Milestone delay	Predicted delay in days at each project milestone	Delay flag	RED (>21 days) / AMBER (7–21 days) / GREEN (<7 days)
Risk score	Schedule risk score per work package (0 to 100)	Weather risk	Monsoon and weather risk overlay by month and city
Critical path	Activities most likely to slip on the critical path	Float advice	Float reallocation and resource buffer recommendation

Crack Intelligence

CRACK INTELLIGENCE · ML MODEL

An AI assistant and recommendation system that analyses structural drawings, approved mix designs, and planned construction processes before any concrete is mixed. It predicts crack occurrence, type and severity, root cause, and remediation cost - then recommends specific pre-pour adjustments to mix design, curing duration, or formwork schedule to prevent the defect before it happens.

Input Dataset

— key fields per data section

Element Identity & Geometry	Structural Configuration	Material & Mix Design	Process & Loading	4 Predictions (C-P1→C-P4)
- Project / Building / Floor ID - Element ID & type (beam/slab/column) - Span / Depth / Width (mm) - Span-to-depth ratio (derived) - Slab thickness & aspect ratio - Pour volume m³ (derived)	- Support condition - Re-entrant corner / joint present - Distance to nearest joint (mm) - Opening proximity (mm) - Cantilever length (mm)	- Concrete grade (M20–M40) - Water-cement ratio (design W/C) - Cement type & admixture - Target slump & max agg. size - Designed cover to rebar (mm)	- Pour month (season proxy) - Curing method + duration (days) - Formwork removal age (days) - Dead + live load (kN/m²) - Early loading before 28 days?	- C-P1: Crack occurrence probability - C-P2: Type & severity (hairline→wide) - C-P3: Root cause classification - C-P4: Remediation cost (INR)

ML Model Outputs

C-P1 Probability	Crack probability score per element (0–100%) - HIGH / MEDIUM / LOW flag	C-P3 Root cause	Poor curing / high W/C / low cover / early load - specific spec gap called out
C-P2 Type	Flexural / Shrinkage / Settlement / Shear - IS 456 width compliance	C-P4 Cost (INR)	Predicted remediation cost: epoxy / stitching / breakout - scaled by severity

Honeycomb Intelligence

HONEYCOMB INTELLIGENCE · ML
MODEL

An AI assistant and recommendation system that analyses the Bar Bending Schedule, approved mix design, and planned casting method before any concrete is placed. It predicts void occurrence, location zone, severity grade, root cause from the casting process, and remediation cost - then recommends specific adjustments to vibration spacing, lift height, or target slump to eliminate the honeycomb risk before the pour date.

Input Dataset

— key fields per data section

Element Identity & Geometry	Rebar Density (BBS)	Mix Workability	Casting Method	4 Predictions (H-P1→H-P4)
- Project / Building / Floor ID - Element ID & type (column/wall/beam) - Element & pour height (mm) - Section width × depth (mm) - Rebar congestion index (derived) - Lift height per layer (mm)	- Main bar dia. + count in section - Link dia. + spacing (mm) - Lap splice location height (mm) - Mechanical couplers / bundled bars? - Clear distance below lap splice (mm)	- Concrete grade Design slump (mm) - Water-cement ratio - Admixture type + dosage (% bwc) - Cement content (kg/m³) - Max aggregate size SCC? (0/1)	- Vibrator type & needle dia. (mm) - Planned insertion spacing (mm) - Pour rate (m³/hr) - Lift height per layer (mm) - Pour month & contractor grade	- H-P1: Void occurrence probability - H-P2: Severity grade (1→3) - H-P3: Root cause classification - H-P4: Remediation cost (INR)

ML Model Outputs

H-P1 Probability	Honeycomb probability score per element (0–100%) - HIGH / MEDIUM / LOW flag	H-P3 Root cause	Inadequate vibration / over-height lift / rapid pour / low slump - % over spec
H-P2 Severity	Grade 1 (surface) / Grade 2 (shallow) / Grade 3 (deep void, rebar exposed)	H-P4 Cost (INR)	Low-pressure / high-pressure injection / breakout - IS 516 core test count

BOQ Deviation Intelligence

BOQ DEVIATION · ML MODEL

An AI assistant and recommendation engine that predicts BOQ quantity, rate, and variation order deviations across every line item - and prescribes specific commercial actions - turning the tendered BOQ into a live financial risk dashboard, updated monthly.

Input Dataset

— key fields per data section

Project & BOQ Identity	Drawings & Scope	Variation & Change Log	Cost & EVM Data	5 Predictions (B1→B5)
- Project ID / Contract No. - Client & contract type (EPC/LS) - Total contract value (INR Cr) - No. of BOQ line items at award - Trade package / section - Tendered Q, rate & item value	- Drawing revision at tender stage - IFC issue date & revision count - Design completion % at tender - Pre-award RFIs raised (count) - Provisional sums & PC items (INR) - BIM / IFC model available?	- Total VOs issued (count & INR Cr) - VO type: client / design / unforeseen - Approved vs pending VO value - VO cumulative run rate (INR/month) - Design revision linked to VO	- Actual cost incurred to date (INR Cr) - Earned value (EV) at measurement date - CPI and SPI at measurement date - Cost-to-complete by trade (INR Cr) - Pending claims / counterclaims (INR)	- B1: Item-level quantity deviation (%) - B2: Rate deviation vs tendered - B3: VO count & value at completion - B4: Final account forecast (INR Cr) - B5: Risk-weighted P50/P80/P90 range

ML Model Outputs

B1 Item deviation	Predicted Q_final vs Q_BOQ per line item (% and units) — top 20 risk items	B4 Final account	Predicted final account (INR Cr) — EAC, VAC, deviation from BAC by trade
B2 Rate variance	Predicted rate deviation per item (INR/UOM) — PAC-eligible split	B5 Risk scenario	P50 / P80 / P90 final account range — contingency adequacy vs required
B3 VO forecast	Predicted VO count and cumulative value at completion — run rate trajectory		

Defect Volume Intelligence

DEFECT VOLUME · ML MODEL

An AI assistant and recommendation system that analyses trade package history, subcontractor records, construction activity, material logs, and weather data - updated floor by floor as the project progresses. It predicts defect count, type distribution, severity grade, root cause, and rework cost one cycle ahead - then recommends targeted pre-emptive actions such as increased inspection hold points, vibration density adjustments, or subcontractor interventions - before the next floor begins.

Input Dataset

— key fields per data section

Project & Trade Profile	Activity & Progress	Material & Site	Historical QA/QC	5 Predictions (P1→P5)
- Project type, GFA & floors - Structural system & contract type - Subcontractor class (A/B/C) - Past defect rate (defects/floor) - Workforce size & skill ratio - Site engineer experience (yrs)	- Activity / work package name - Construction stage & SPI - Concurrent activities running - Prior rework on same element - QC hold point compliance rate (%) - Third-party inspection involved?	- Material type, grade & source - Approved vs non-approved supplier - Delivery variance (days) - Test certificate status (P/F) - Site storage condition - Weather / non-productive days	- Defects recorded to date (count) - Defect rate per floor - current project - Top 3 defect types on this project - Defect closure rate (% within 7 days) - Portfolio avg for project type & city	- P1: Total defect count per floor - P2: Type split (structural/MEP/finishes) - P3: Severity grade distribution - P4: Root cause classification - P5: Rework cost & programme impact

ML Model Outputs

P1 Volume	Predicted defect count per floor - HIGH / MEDIUM / LOW vs project benchmark	P4 Root cause	Process / Material / Supervision / Weather / Design - specific gap called out
P2 Types	Predicted split: Structural / MEP / Finishes / Facade - top 3 defect types	P5 Rework cost	Predicted rework INR per floor - programme days lost - batch saving opportunity
P3 Severity	Minor / Moderate / Major / Critical % - latent defect risk score		

Material Waste Intelligence

MATERIAL WASTE · ML MODEL

An AI assistant and recommendation system that analyses project parameters, site conditions, contractor history, and storage conditions - then predicts material waste risk across all material types before procurement is finalised. It recommends specific procurement adjustments, storage upgrades, and process interventions to reduce waste before orders are placed. An AI assistant answers any question against project data in real time.

Input Dataset

— key fields per data section

Project Profile	Site & Contractor	Material Records	Design & Defect Signals	Historical Context
- Project type (residential / commercial / industrial) - Gross floor area (m²) and number of floors - Structural system (RCC / steel / composite) - Planned project duration and city / geography	- Contractor class and past performance history - Site manager assigned to project - Storage condition (enclosed / semi-covered / open yard) - Construction start month (season proxy)	- Material type (structural / MEP / finishes / joinery / waterproofing) - Planned procurement quantity per material - Unit of measurement (m³ / m² / kg / nos / litre) - Material category and supplier tier	- Design revision count (actual or predicted) - Number of QA/QC defects recorded on project - Rework instances on similar elements - Weather exposure and non-productive days	- Past waste rates on similar project types - Contractor waste history across L&T portfolio - City-level material wastage benchmarks - Seasonal weather risk by location and month

ML Model Outputs

Waste rate forecast	Predicted waste rate (%) per material type — HIGH / MEDIUM / LOW risk banding before procurement	Defect-waste linkage	Predicted defect count and its downstream impact on material wastage across trades
Wasted quantity	Predicted wasted quantity in native UOM per material — enables right-sizing of purchase orders	Procurement actions	Recommended order quantity adjustments, storage upgrades, and contractor-level interventions
Design revision risk	Predicted design revision count from project parameters — flags rework-driven waste risk early	AI assistant	Plain-English Q&A on any waste, material, project, or site query — analytics and knowledge paths

Material Waste Intelligence

MATERIAL WASTE · ML MODEL

4 ML MODELS · 110 MATERIALS

This working prototype validates the intelligence model using synthetic L&T-grade data - predicting material waste rate and quantity per material type, recommending targeted procurement and storage actions, and surfacing insights through a built-in AI assistant. Every output on this slide is a live prediction from the model, not a manually curated estimate.

Sample Project Input

Type:

Residential

GFA:

12,000 m² · 12 Floors

Structure:

RCC Frame

Location:

Mumbai

Contractor:

Class C

Storage:

Poor (Open Yard)

Season:

June (Monsoon onset)

4-Model Prediction Cascade

M1

Design Revision Predictor

Features: Type · GFA · Floors · Structure · Location ·

→ 1.2 Design Revisions

M2

Defect Count Predictor

Features: All M1 inputs + Design Revisions + Rainy

→ 12 Predicted Defects

M3

Waste Rate Predictor (per material)

Features: All M2 inputs + Material Type + Planned

→ Waste Rate % per material

M4

Wasted Quantity Predictor

Features: Same as M3

→ Wasted Qty in native UOM

Dataset Schema · 17 Columns

50,015

Rows

110

Materials

21

Cities

- Project_ID, Project_Type
- GFA_m2, Num_Floors
- Structure_Type
- Material_Type
- Planned_Quantity, Unit
- Storage_Condition
- Contractor_Class
- Design_Revisions
- Location
- Rainy_Days_Share_%
- Site_Manager_Name
- Number_of_Defects
- Wastage_Rate_% ← target

Predicted Outputs — Sample Project: Residential · Mumbai · 12,000 m² · RCC · Class C · Poor Storage

12

Predicted Defects

QA/QC defect count

1.2

Design Revisions

revision cycles forecast

21.7%

Peak Waste Rate

highest material — HIGH RISK

Top 5 Highest-Risk Materials (from full prediction across all material types)

1. Timber Batten 2x2	21.67%	HIGH
2. Plywood 12mm	21.64%	HIGH
3. Paint Roller 9-inch	21.63%	HIGH
4. Nails 50mm	21.60%	HIGH
5. Glass Panel 12mm	21.43%	HIGH